CB.SC.U4CYS23029

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WEB SCRAPER USING PYTHON

scraper.py

C: > CLASS DOCUMENTS > SEM-7 > DS AND CC > DOCKER > LAB-DOCKER > python_web_scrapper > scraper.py

```python
import os
import requests
from bs4 import BeautifulSoup

# Read TARGET_URL from the environment
target_url = os.getenv("TARGET_URL", "https://news.ycombinator.com")

try:
    # Fetch the webpage
    response = requests.get(target_url)
    response.raise_for_status()

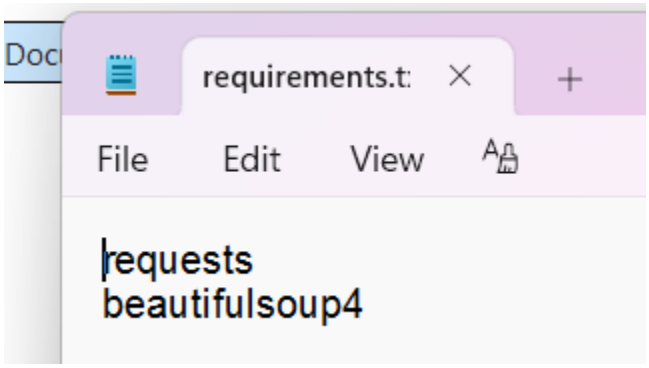
    # Parse the HTML
    soup = BeautifulSoup(response.text, "html.parser")

    # Find all h1 and h2 headings
    titles = soup.find_all(["h1", "h2"])

    print(f"\nScraping: {target_url}\n")

    if titles:
        for i, title in enumerate(titles, 1):
            print(f"{i}. {title.get_text(strip=True)}")
    else:
        print("No h1 or h2 headings found.")

except Exception as e:
    print(f"Error fetching or parsing the URL: {e}")
```



Dockerfile

File Edit View H1

```
# Use official Python image
FROM python:3.10-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install required packages
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY scraper.py .

# Set environment variable
ENV TARGET_URL=https://news.ycombinator.com

# Expose port 8080
EXPOSE 8080

# Run the scraper
CMD ["python", "scraper.py"]
```

Building the image:

```
C:\Users\nukal>cd "C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper"

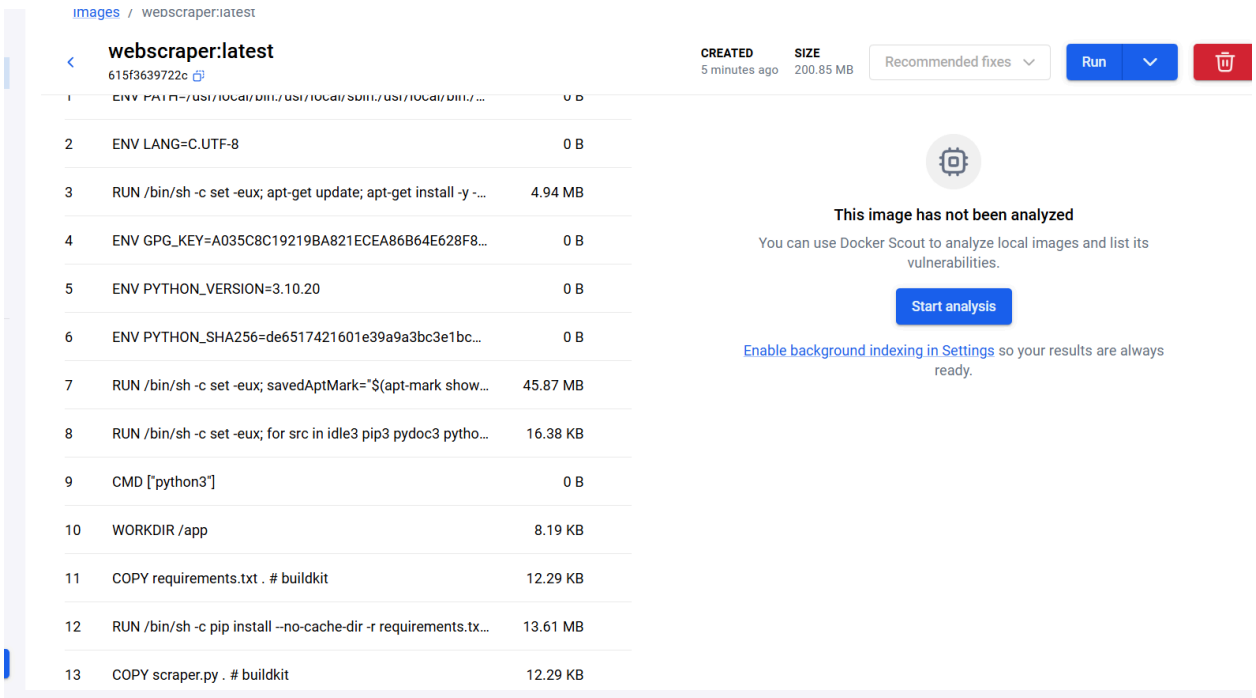
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker build -t webscraper .
[+] Building 270.8s (7/10)                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                       0.0s
 => => transferring dockerfile: 473B                                                                                       0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                                                       37.2s
 => [auth] library/python:pull token for registry-1.docker.io                                                              0.0s
 => [internal] load .dockerignore                                                                                          0.0s
 => => transferring context: 2B                                                                                            0.0s
 => [1/5] FROM docker.io/library/python:3.10-slim@sha256:123f8aa87ae9112f0fccc9b9cc342c79e2f8f1e2b2e10395cd6b2453d1b42bd1  233.5s
 => => resolve docker.io/library/python:3.10-slim@sha256:123f8aa87ae9112f0fccc9b9cc342c79e2f8f1e2b2e10395cd6b2453d1b42bd1   0.0s
 => => sha256:acf9e06c2d45a94472a7d893455c898eff1f037b1c86b3e32f2dc67c57418da7 251B / 251B                                 6.0s
 => => sha256:2f7d10ea5bd937f48518553448d94b63ca5cc4a921cf96a374dee6d36b17c03c 13.63MB / 13.88MB                         233.4s
 => => sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1 1.29MB / 1.29MB                            23.2s
 => => sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdff2c05c6a053 10.49MB / 29.78MB                         233.4s
 => [internal] load build context                                                                                          0.1s
 => => transferring context: 848B                                                                                          0.0s
 => CANCELED [2/5] WORKDIR /app                                                                                            0.0s
ERROR: failed to build: failed to solve: Canceled: context canceled
```

```
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker build -t webscraper .
[+] Building 52.0s (10/10) FINISHED                                                                         docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                       0.0s
 => => transferring dockerfile: 473B                                                                                       0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                                                        1.1s
 => [internal] load .dockerignore                                                                                          0.0s
 => => transferring context: 2B                                                                                            0.0s
 => [1/5] FROM docker.io/library/python:3.10-slim@sha256:123f8aa87ae9112f0fccc9b9cc342c79e2f8f1e2b2e10395cd6b2453d1b42bd1   24.5s
 => => resolve docker.io/library/python:3.10-slim@sha256:123f8aa87ae9112f0fccc9b9cc342c79e2f8f1e2b2e10395cd6b2453d1b42bd1   0.0s
 => => sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdff2c05c6a053 29.78MB / 29.78MB                           23.6s
 => => extracting sha256:062e450697faa5f02a3a74eba9864ee4d79bc9cfbd65769fc6cdff2c05c6a053                                  0.5s
 => => extracting sha256:60e855e0dd9540cc6da6057297f1ab688e602e650a6075fc14da2c2f9c8884d1                                  0.1s
 => => extracting sha256:2f7d10ea5bd937f48518553448d94b63ca5cc4a921cf96a374dee6d36b17c03c                                  0.3s
 => => extracting sha256:acf9e06c2d45a94472a7d893455c898eff1f037b1c86b3e32f2dc67c57418da7                                  0.0s
 => [internal] load build context                                                                                          0.0s
 => => transferring context: 67B                                                                                           0.0s
 => [2/5] WORKDIR /app                                                                                                     0.1s
 => [3/5] COPY requirements.txt .                                                                                          0.0s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                              24.5s
 => [5/5] COPY scraper.py .                                                                                                0.2s
 => exporting to image                                                                                                     1.4s
 => => exporting layers                                                                                                    0.9s
 => => exporting manifest sha256:c0739fc02f7373bdde059884dac26179e1e60d7f069a7604e8a6143840115fbb                           0.0s
 => => exporting config sha256:dd74553f1039e5956dded4cc09c6081766992e78825d42d3230d12c1e40d9972                            0.0s
 => => exporting attestation manifest sha256:23282b502fd13b46d3a3e0fe2e78e1e88174809f0284b29a6f030be07e3aad59              0.0s
 => => exporting manifest list sha256:b3ba8fae60fb052a81d5592481c202014bda5bf06f567853d1181d98b0cc9252                     0.0s
 => => naming to docker.io/library/webscraper:latest                                                                       0.0s
 => => unpacking to docker.io/library/webscraper:latest                                                                    0.3s

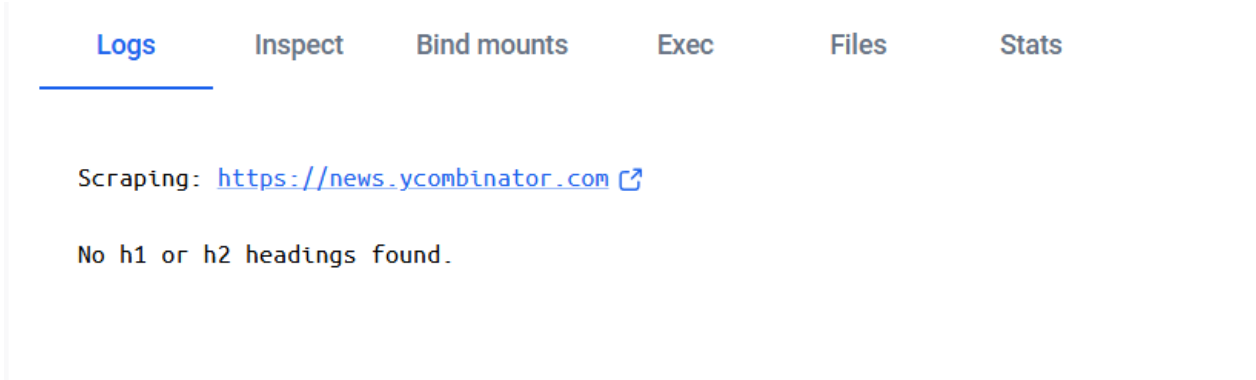
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/erk7rq9g1zo6aanoffb44o4pv
```

```
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker images

IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
akshayn789/filelogsdockerimage:latest          3955e7217b57        203MB         49.6MB
akshayn789/filesharingdockerimage:latest       c98667f326ca        210MB         51.2MB
akshayn789/firstdockerimage:latest             71fe7f78b3cb       6.84MB         2.23MB
akshayn789/flaskappdockerimage:latest          fad96ec1e64a        259MB         70.5MB
akshayn789/portexpose:latest                   d0f6c032ae5c       92.5MB           26MB
demoimage:latest                               71fe7f78b3cb       6.84MB         2.23MB
filesharingdockerimage:latest                  c98667f326ca        210MB         51.2MB
flaskappdockerimage:latest                     fad96ec1e64a        259MB         70.5MB
html-site:latest                               d0f6c032ae5c       92.5MB           26MB
nginx:latest                                   ec4ed8b5299e        241MB           66MB
portexpose:latest                              d0f6c032ae5c       92.5MB           26MB
timelogsdockerimage:latest                     3955e7217b57        203MB         49.6MB
webscraper:latest                              615f3639722c        201MB         48.9MB
```



The logs after the running the container:



```
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker run -e TARGET_URL=https://example.com webscraper
```

Changed the scraper code:

```
titles = soup.select(".titleline a")
```

```
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker build -t webscraper .
[+] Building 5.2s (11/11) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 473B
 => [internal] load metadata for docker.io/library/python:3.10-slim
 => [auth] library/python:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/5] FROM docker.io/library/python:3.10-slim@sha256:123f8aa87ae9112f0fccc9b9cc342c79e2f8f1e2b2e10395cd6b2453d1b42bd1
 => => resolve docker.io/library/python:3.10-slim@sha256:123f8aa87ae9112f0fccc9b9cc342c79e2f8f1e2b2e10395cd6b2453d1b42bd1
 => [internal] load build context
 => => transferring context: 816B
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY scraper.py .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:3de1a62593bf83f74ce25b9f612690dbb5099f922f2cb9316cf2dc1568a83ee5
 => => exporting config sha256:4cb3496a1fa96a716d20ac65c4adb05e6af8f22a88e09cd3d936256e32016b6c
 => => exporting attestation manifest sha256:5291d6e96cd92bd5de14dd82b4b2d558c651ab92be2eda695be19ffa401146a8
 => => exporting manifest list sha256:227ccd83b69afa359f259d16bee96c64e71e2845ae444554faecc4b9391d971a
 => => naming to docker.io/library/webscraper:latest
 => => unpacking to docker.io/library/webscraper:latest

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/lzt7bpplaovuika4sogz3yzp6
```

Rebuilt the image

Got the info now:

```
Scraping: https://news.ycombinator.com

1. The lost joy of music piracy
2. pigeonsandplanes.com
3. Inkling: Our Open-Weights Model
4. thinkingmachines.ai
5. If you want to create a button from scratch, you must first create the universe
6. madcampos.dev
7. What's the story behind the names of Cloudflare's name servers? (2013)
8. cloudflare.com
9. Teardown: A Generic 7-Port USB 3.0 Hub That Wasn't
10. goughlui.com
11. 1,300 Beautiful Wildlife Illustrations from the 19th Century Now Restored
12. openculture.com
13. Grok Build is open source
14. github.com/xai-org
15. SQLite should have (Rust-style) editions
16. mort.coffee
17. Governments, companies, nonprofits should invest in free, open source AI [pdf]
18. siegelendowment.org
19. Stripe and Advent have made a joint offer to acquire PayPal - sources
20. reuters.com
21. Bluesky Trademarks ATProto
22. atproto.com
23. Reynard: A real Firefox web browser for iOS 13 or later
```

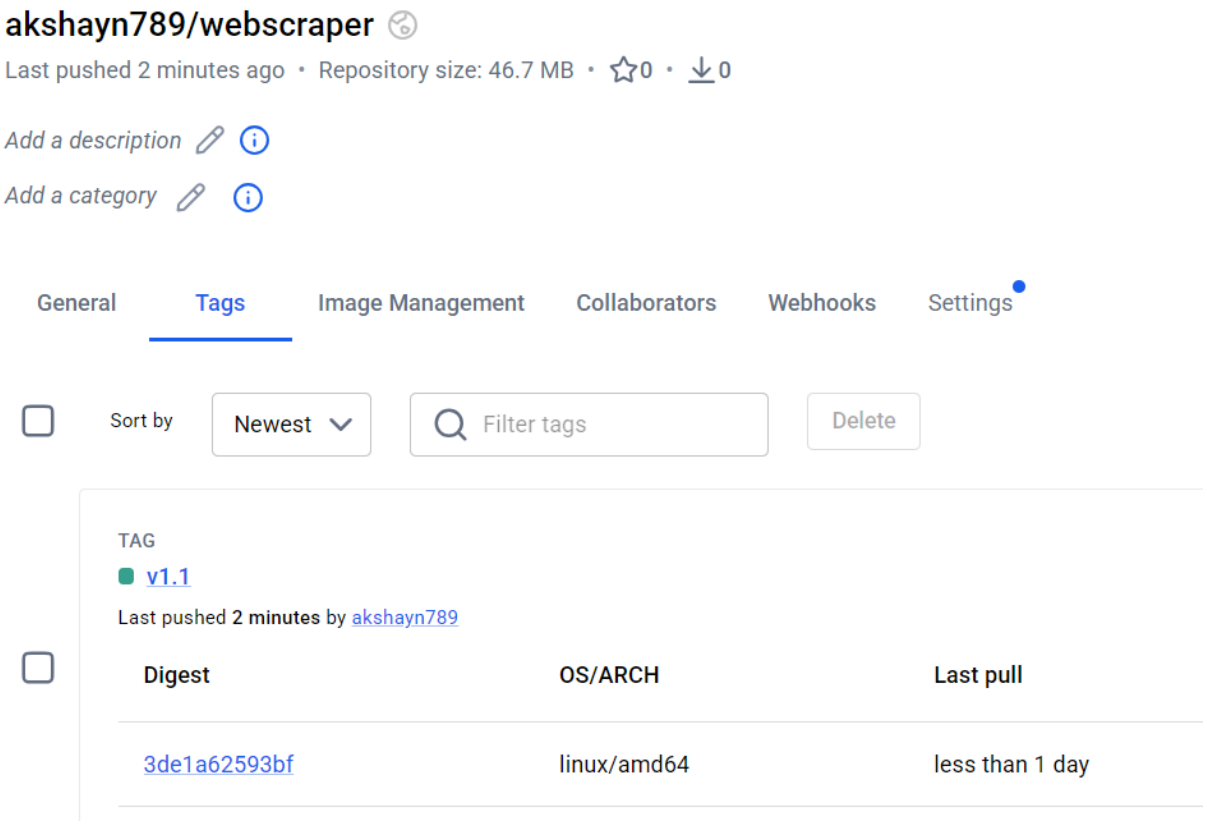
Tagging the image

```
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker tag webscraper akshayn789/webscraper:v1.1
```

Pushing into a repo

```
C:\CLASS DOCUMENTS\SEM-7\DS AND CC\DOCKER\LAB-DOCKER\python_web_scrapper>docker push akshayn789/webscraper:v1.1
The push refers to repository [docker.io/akshayn789/webscraper]
e1a1d7da6446: Pushed
2ebd317aa01f: Pushed
393d40da83c7: Pushed
acf9e06c2d45: Pushed
60e855e0dd95: Pushing [==================================================>]  1.293MB/1.293MB
bf7da3f0fea2: Pushed
34dac9eda3bc: Pushing [=============>                                     ]  1.049MB/3.984MB
2f7d10ea5bd9: Pushing [===>                                               ]  1.049MB/13.88MB
062e450697fa: Pushing [=>                                                 ]  1.049MB/29.78MB
```

```
v1.1: digest: sha256:227ccd83b69afa359f259d16bee96c64e71e2845ae444554faecc4b9391d971a size: 856
```



Link: https://hub.docker.com/r/akshayn789/webscraper